

Intro to Data Visualization

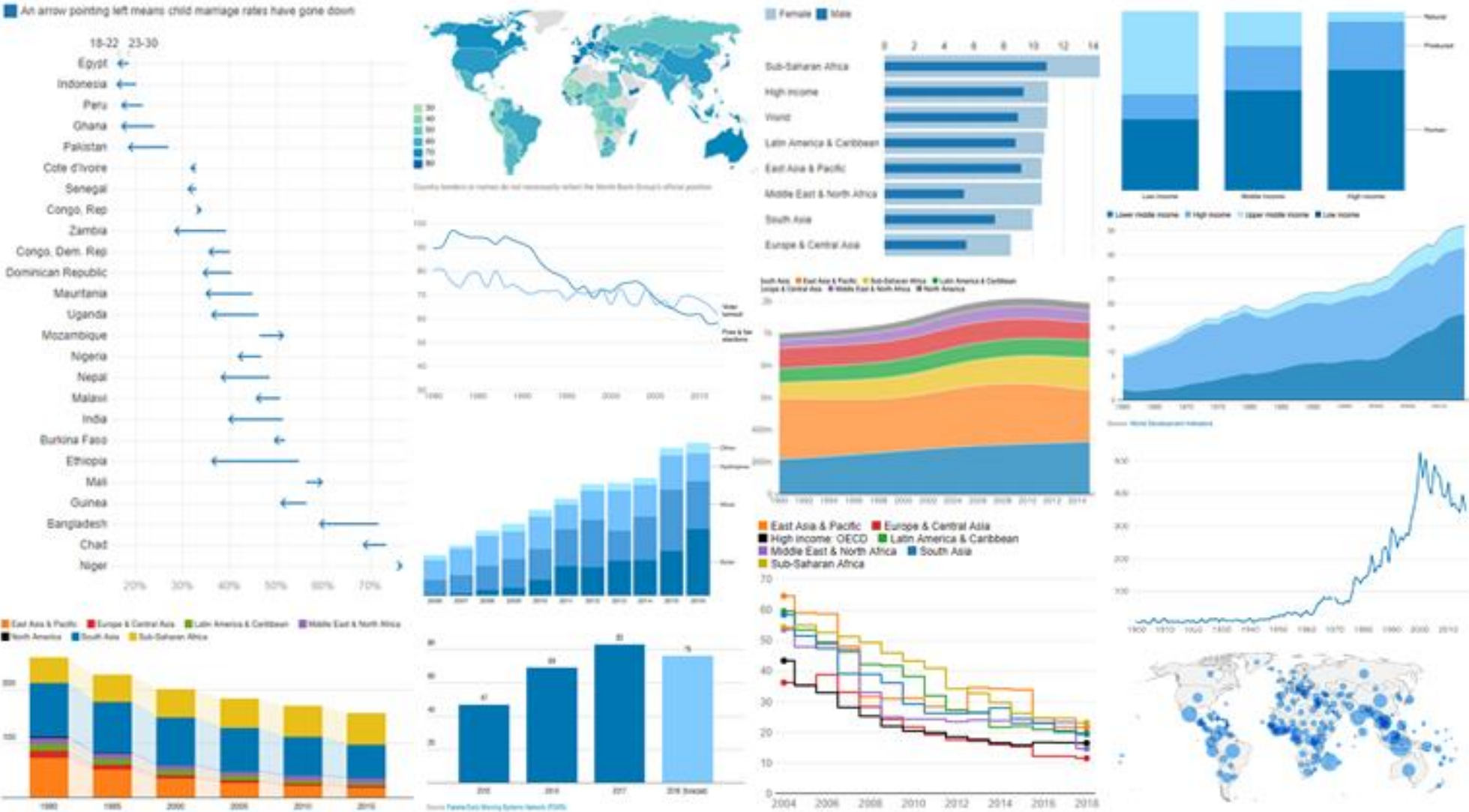
Data manipulation and plotting

Big data



Insights, predictions, products

Data visualization



Modules

Pandas - reading data

Matplotlib - plotting data

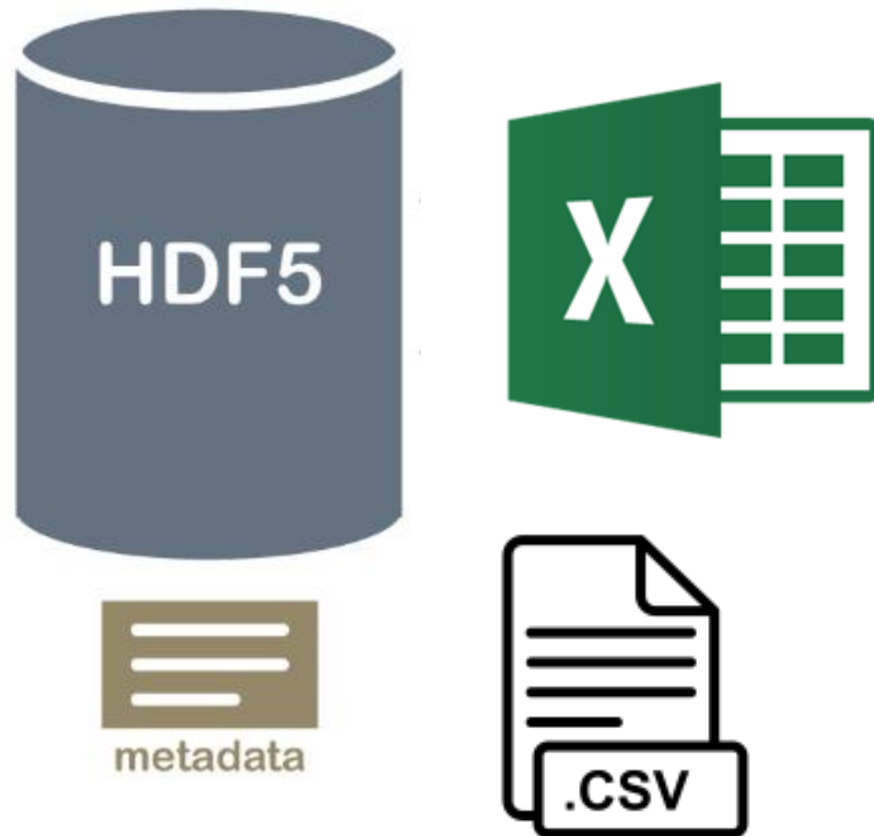
What is **Pandas** ?

- *Pandas* is an open source, BSD-licensed library providing high-performance, easy-to-use data structures and data analysis tools for the Python programming language.



What does Pandas can do ?

Reading data



Viewing data

	gender	race/ethnicity	parental level of education	lunch \
0	female	group C	some high school	standard
1	female	group D	high school	standard
2	female	group E	high school	standard
3	male	group C	high school	standard
4	male	group C	associate's degree	standard
5	female	group D	associate's degree	standard
6	male	group D	high school	standard
7	female	group E	some high school	free/reduced
8	female	group D	associate's degree	free/reduced
9	female	group D	some college	standard
10	female	group D	some high school	free/reduced
11	female	group B	high school	free/reduced
12	male	group C	some college	standard
13	female	group B	some college	standard
14	male	group C	high school	standard
15	female	group B	some college	standard
16	female	group E	some college	free/reduced
17	male	group E	some college	standard
18	female	group C	associate's degree	standard
19	female	group E	high school	standard
20	male	group B	some high school	free/reduced
21	male	group E	master's degree	free/reduced
22	male	group C	some high school	standard
23	male	group C	some high school	standard


```
{
  "about": "Sunlight Foundation uses cutting-edge technology and ideas to make
  "category": "Non-governmental organization (ngo)",
  "category_list": [
    {
      "id": "2235",
      "name": "Non-Governmental Organization (NGO)"
    }
  ],
  "checkins": 136,
  "description": "POSTING GUIDELINES:\n\nThe Sunlight Foundation p
ansparency and general issues related to open data. Individuals po
```

json

```
id,name,released_on,price,created_at,updated_at
24,1000 Piece Jigsaw Puzzle,2012-07-03,14.99,2012-07-09 16:50:49 UTC,2012-07-09 16:50:49 UTC
30,360° Protractor,2012-05-03,3.99,2012-07-09 16:50:49 UTC,2012-07-09 16:50:49 UTC
17,7 Wonders,2012-04-21,28.75,2012-07-09 16:50:49 UTC,2012-07-09 16:50:49 UTC
13,Acoustic Guitar,2012-06-06,1025.0,2012-07-09 16:50:49 UTC,2012-07-09 16:50:49 UTC
15,Agricola,2012-05-22,45.99,2012-07-09 16:50:49 UTC,2012-07-09 16:50:49 UTC
22,Answer to Everything,2012-07-03,42.0,2012-07-09 16:50:49 UTC,2012-07-09 16:50:49 UTC
23,Box Kite,2012-05-19,63.0,2012-07-09 16:50:49 UTC,2012-07-09 16:50:49 UTC
29,CanCan Music Record,2012-05-09,2.99,2012-07-09 16:50:49 UTC,2012-07-09 16:50:49 UTC
12,Chocolate Pie,2012-04-12,3.14,2012-07-09 16:50:49 UTC,2012-07-09 16:50:49 UTC
9,Dog Toy Bone,2012-06-13,2.99,2012-07-09 16:50:49 UTC,2012-07-09 16:50:49 UTC
11,Flux Capacitor,2012-06-01,19.55,2012-07-09 16:50:49 UTC,2012-07-09 16:50:49 UTC
6,Game Console,2012-06-06,299.95,2012-07-09 16:50:49 UTC,2012-07-09 16:50:49 UTC
10,Heated Blanket,2012-07-19,27.95,2012-07-09 16:50:49 UTC,2012-07-09 16:50:49 UTC
19,Knights of Catan,2012-06-10,19.95,2012-07-09 16:50:49 UTC,2012-07-09 16:50:49 UTC
8,Lawn Chair,2012-05-29,34.99,2012-07-09 16:50:49 UTC,2012-07-09 16:50:49 UTC
21,Millennium Falcon,2012-04-10,3597200.0,2012-07-09 16:50:49 UTC,2012-07-09 16:50:49 UTC
14,Model Enterprise,2012-04-18,27.99,2012-07-09 16:50:49 UTC,2012-07-09 16:50:49 UTC
28,Model Train Rails,2012-06-30,45.0,2012-07-09 16:50:49 UTC,2012-07-09 16:50:49 UTC
3,Oak Coffee Table,2012-07-08,223.99,2012-07-09 16:50:49 UTC,2012-07-09 16:50:49 UTC
5,Oh's Cereal,2012-04-17,3.95,2012-07-09 16:50:49 UTC,2012-07-09 16:50:49 UTC
```

CSV

How to install Pandas

- Conda command: `conda install -c anaconda pandas`
- Pip command: `pip install pandas`

Some commands of Pandas

1. How to import Pandas.

Command : `import pandas as pd`

2. Reading a csv file.

Command : `pd.read_csv(filepath_or_buffer)`

filepath_or_buffer (String) = Path of a file.

Example: `df = pd.read_csv("exams_100.csv")`

Some commands of Pandas

3. Viewing the top **n** rows of a file

Command : `df.head(n)`

n (int) = number of rows to select. (Default is **5**)

4. Viewing the bottom **n** rows of a file

Command : `df.tail(n)`

n (int) = number of rows to select. (Default is **5**)

Example of reading data by pandas

Import pandas as pd

```
data = pd.read_csv('data.csv')
```

```
print(data)
```



	gender	group	level of education	test preparation course	math score \
0	female	group C	some high school	none	67
1	female	group D	high school	completed	66
2	female	group E	high school	none	76
3	male	group C	high school	completed	70
4	male	group C	associate's degree	none	56
	reading score		writing score		
0	65		69		
1	75		78		
2	74		75		
3	76		67		
4	49		49		

How to get data after reading it

We can look inside each column by **its name**

Example :

gender = data['gender']



A diagram with two red labels, 'Dict' and 'Key', positioned above the code. A grey arrow points from 'Dict' to the variable 'data', and another grey arrow points from 'Key' to the string 'gender' within the brackets.

Returns : DataFrame

What is a DataFrame ? : a tabular data (with rows and columns).

Convert to list by mylist = list(gender)

What is Matplotlib

- Matplotlib is a Python 2D plotting library



What does Matplotlib can do ?

- Advantages of Matplotlib library
 - Large community
 - Many plot types supported
 - Easy to use with python

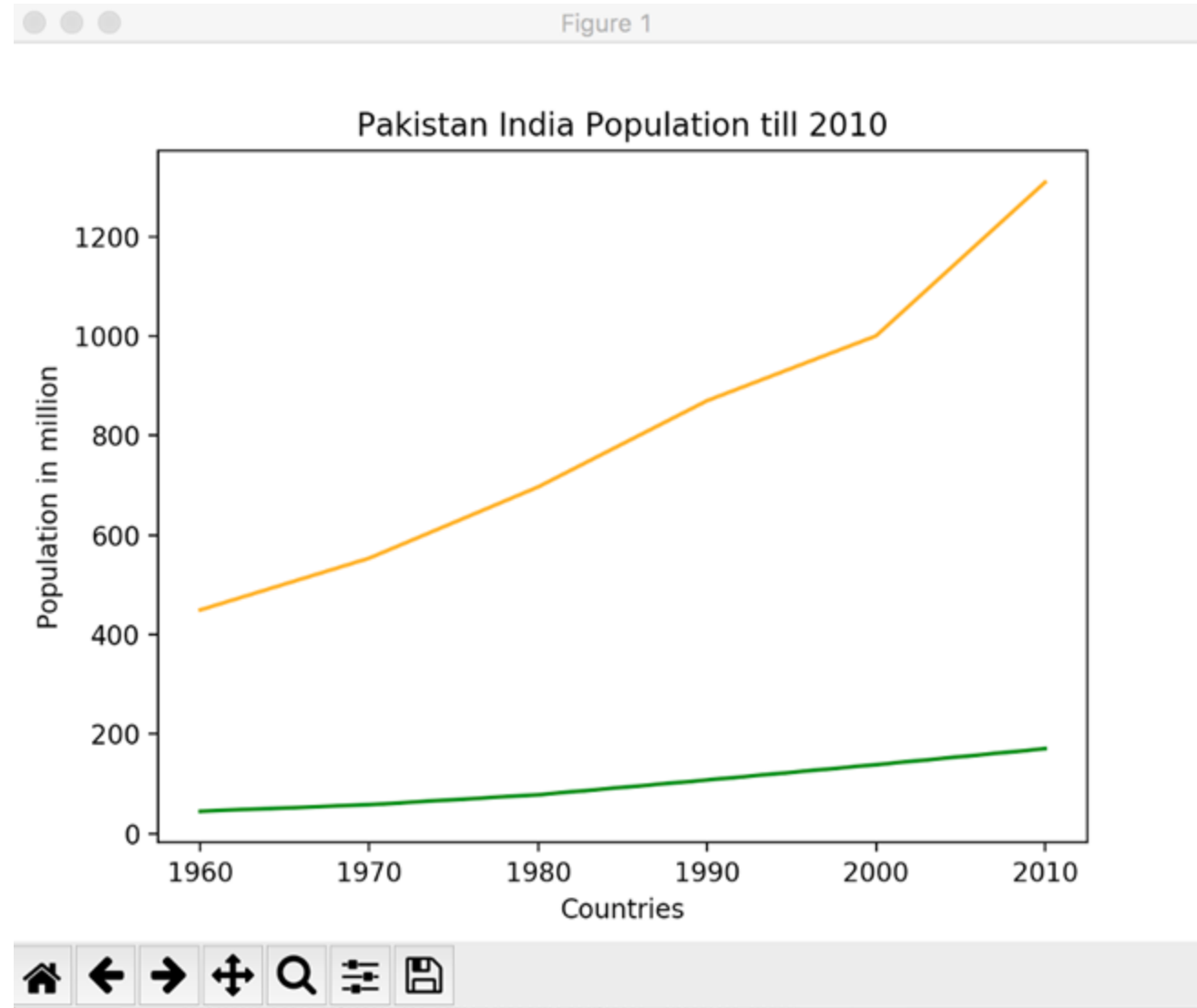
What does Matplotlib can do ?

- Some types of Matplotlib graphs
 - Line graph
 - Bar graph
 - Pie graph
 - etc

Line graph

```
import matplotlib.pyplot as plt

year = [1960, 1970, 1980, 1990, 2000, 2010]
pop_pakistan = [44.91, 58.09, 78.07, 107.7, 138.5, 170.6]
pop_india = [449.48, 553.57, 696.783, 870.133, 1000.4, 1309.1]
plt.plot(year, pop_pakistan, color='g')
plt.plot(year, pop_india, color='orange')
plt.xlabel('Countries')
plt.ylabel('Population in million')
plt.title('Pakistan India Population till 2010')
plt.show()
```

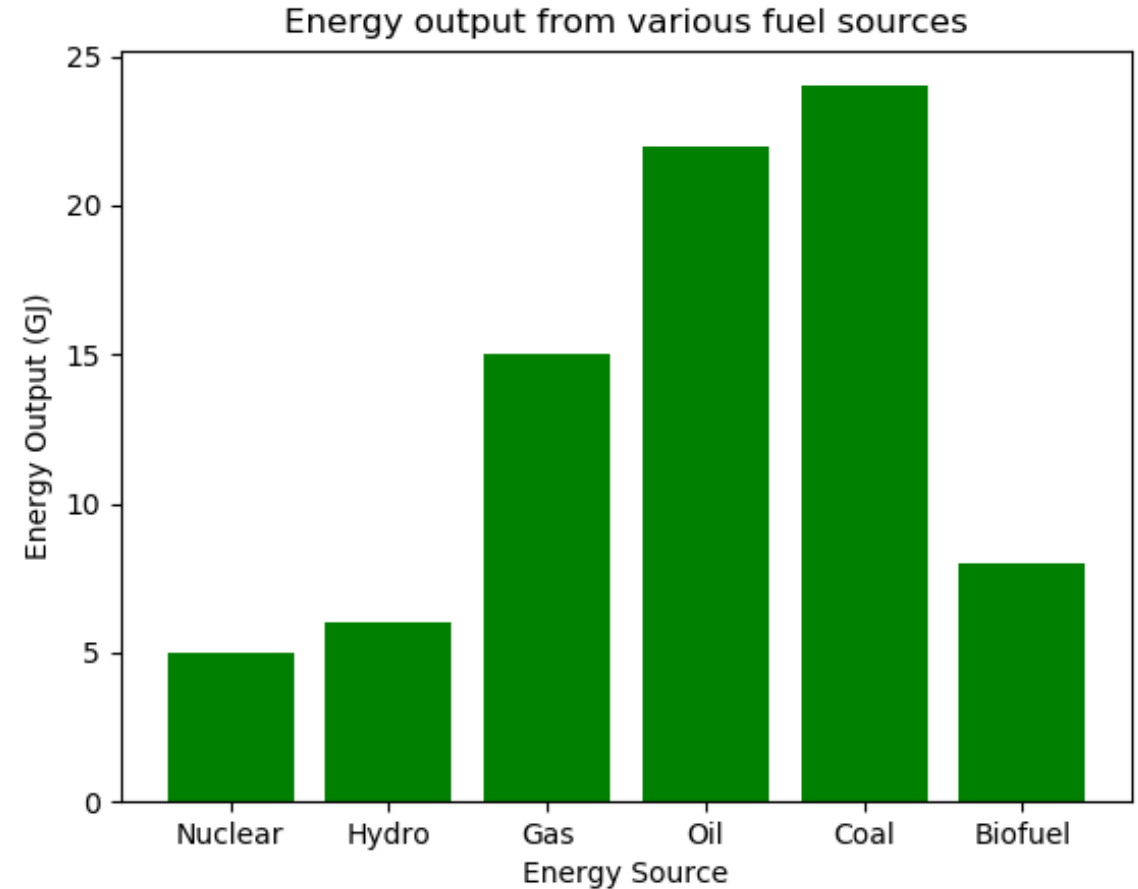


Bar graph

list comprehension!
What does enumerate(x) do?

```
import matplotlib.pyplot as plt

x = ['Nuclear', 'Hydro', 'Gas', 'Oil', 'Coal',
     'Biofuel']
energy = [5, 6, 15, 22, 24, 8]
x_pos = [idx for idx, val in enumerate(x)]
plt.bar(x_pos, energy, color='green')
plt.xlabel("Energy Source")
plt.ylabel("Energy Output (GJ)")
plt.title("Energy output from various fuel
sources")
plt.xticks(x_pos, x)
plt.show()
```



```
mylist = ['a','b','c']
```

```
for item in mylist
```

```
for id, item in enumerate(mylist)
```

Bar graph with multiple plots

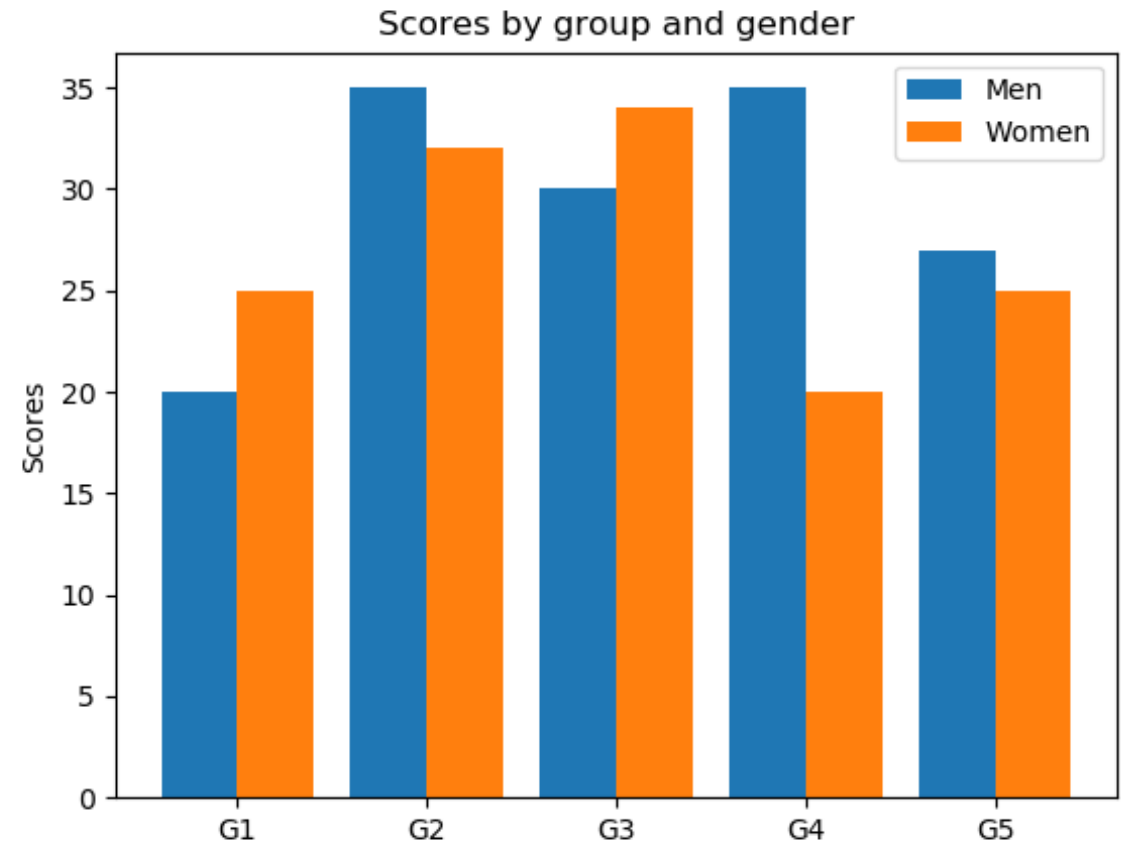
```
N = 5
men_means = (20, 35, 30, 35, 27)
women_means = (25, 32, 34, 20, 25)
width = 0.40

index_men = [idx for idx in range(N)]
index_women = [idx+0.40 for idx in range(N)]
index_number = [idx+0.40 / 2 for idx in range(N)]

plt.bar(index_men, men_means, width, label='Men')
plt.bar(index_women, women_means, width, label='Women')

plt.ylabel('Scores')
plt.title('Scores by group and gender')

plt.xticks(index_number, ('G1', 'G2', 'G3', 'G4', 'G5'))
plt.legend(loc='best')
plt.show()
```



Pie graph

```
import matplotlib.pyplot as plt
```

```
# Data to plot
```

```
labels = 'Python', 'C++', 'Ruby', 'Java'
```

```
sizes = [215, 130, 245, 210]
```

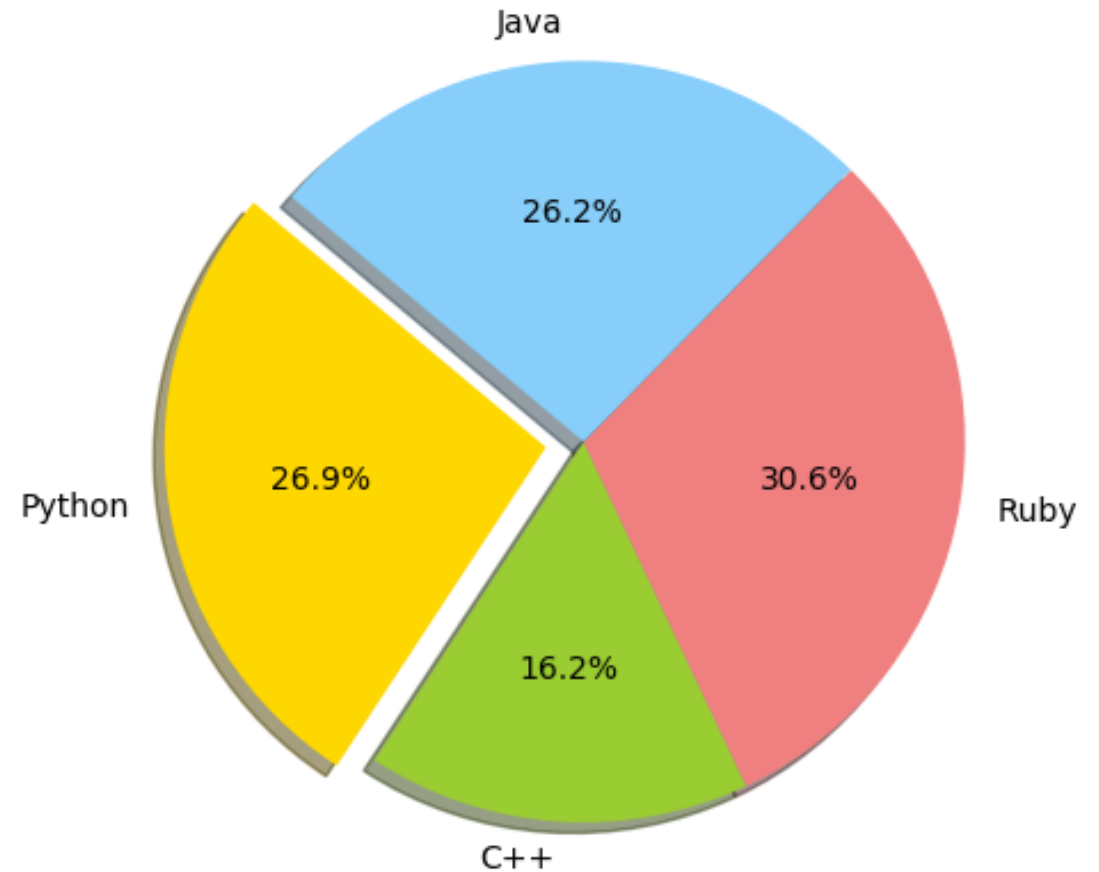
```
colors = ['gold', 'yellowgreen', 'lightcoral',  
          'lightskyblue']
```

```
explode = (0.1, 0, 0, 0) # explode 1st slice
```

```
# Plot
```

```
plt.pie(sizes, explode=explode,  
        labels=labels, colors=colors,  
        autopct='%1.1f%%', shadow=True,  
        startangle=140)
```

```
plt.show()
```



Pie graph calculation

First, put your data into a table (like above), then add up all the values to get a total:

<i>Table: Favorite Type of Movie</i>					
Comedy	Action	Romance	Drama	SciFi	TOTAL
4	5	6	1	4	20

Next, divide each value by the total and multiply by 100 to get a percent:

Comedy	Action	Romance	Drama	SciFi	TOTAL
4	5	6	1	4	20
$\frac{4}{20}$ = 20%	$\frac{5}{20}$ = 25%	$\frac{6}{20}$ = 30%	$\frac{1}{20}$ = 5%	$\frac{4}{20}$ = 20%	100%

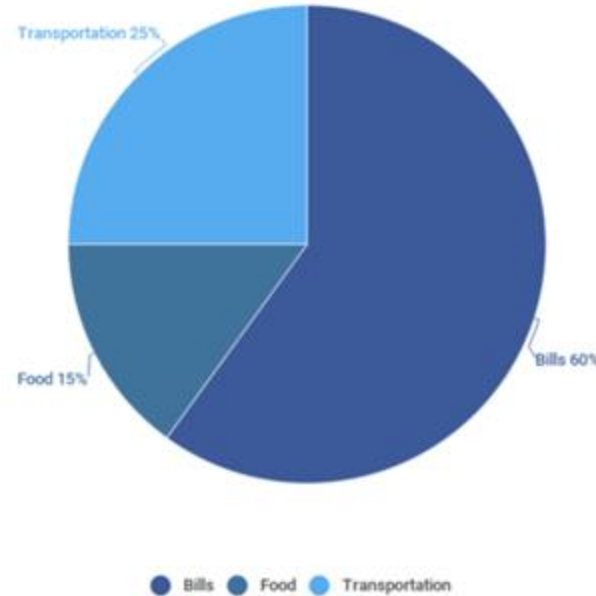
Credit: <https://www.mathsisfun.com/data/pie-charts.html>

Which plot type?

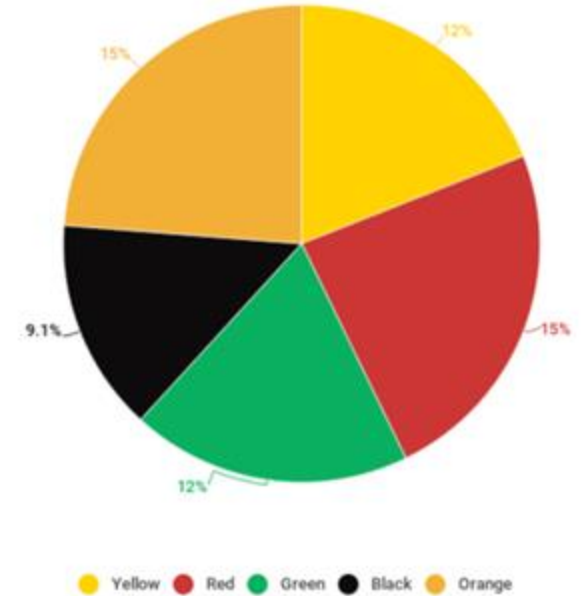
Bar graph: Comparison

Line graph: Trend, time series

Pie Chart - Good Example



Pie Chart - Bad Example



Pie graph: proportion, clear difference, not more than 5 segments